

Elangeshwaran Kannabiran

Ph: +91-90666-28011 | E-mail: elangeshwaran1@gmail.com | Portfolio: elan0705.github.io/Elangeshwaran |

LinkedIn: [elangeshwaran-kannabiran](https://www.linkedin.com/in/elangeshwaran-kannabiran) | GitHub: [elan0705](https://github.com/elan0705)

PROFESSIONAL SUMMARY

Robotics and Biomedical Engineering professional with an MSc in Computational Neuroscience & Cognitive Robotics and a BEng in Medical Electronics Engineering, combining hands-on robotic design with human-centred technology development. Experienced in designing and prototyping 5–8 DoF robotic systems, inverse kinematics, computer vision, sensing, Python, MATLAB, machine learning, data analysis, biomechanics, motion capture, eye-tracking and human behaviour analysis. Particularly interested in humanoid robotics, human–robot interaction, bio-inspired design, assistive technologies, and robotics R&D. Alongside engineering experience, bring capabilities in product development, business analytics, financial analysis, and technology innovation, enabling a multidisciplinary approach to identifying real-world problems, developing technical solutions, and evaluating their practical and commercial potential.

CORE SKILLS

Programming & Robotics: Python, MATLAB, Arduino, Kotlin, Linux, Django (Python web framework), Computer Vision, HTML, CSS, C & Java

2D/3D Design & CAD: Siemens NX, Fusion 360, AutoCAD, SolidEdge 2D, 3D Printing

Business & Analytics: Business Analytics, Financial Management, Git/GitHub, Microsoft Office

EDUCATION

MSc, Computational Neuroscience & Cognitive Robotics *The University of Birmingham — Birmingham, United Kingdom*

BEng, Medical Electronics Engineering *Dayananda Sagar College of Engineering — Bangalore, India*

WORK EXPERIENCE

Biomedical Engineering Intern, Sagar Hospitals *Bangalore, India | Feb 2019 – May 2019*

- Conducted a gap analysis of existing versus required hospital technology to inform equipment planning.
- Analysed equipment quality-control protocols across five hospital departments as a core team member.
- Formulated a preventive-and-corrective maintenance protocol and mentored junior student groups to minimize risk and ensure safety.
- Observed and documented the role and responsibilities of a Biomedical Engineer across multiple medical procedures.

RESEARCH & PROJECTS

Dexterous Robot Arm for Human-Like Grasping of Objects

- Achieved close-to-human motion on a robotic arm by applying human cognitive approaches, identifying models and methods applicable to humanoids and other social robots.
- Investigated human structural anatomy and cognitive approaches relevant to reach-and-grasp tasks.
- Designed and 3D-printed an 8-degree-of-freedom robotic arm with the closest structural and mechanical resemblance to a human arm.
- Developed an inverse-kinematics model integrated with a computer-vision model.
- Established environment connections and evaluated reach-and-grasp task performance on the robot arm.

Project archive: [MSc. Thesis](#)

Telesurgical Bionic Arm

- Designed a cost-effective, wirelessly communicated master–slave bionic arm to support rural healthcare delivery.
- Built a 5-degree-of-freedom robot arm with human resemblance.
- Integrated flex sensors, potentiometers, and an accelerometer on a physician-worn glove as a single-input, multi-output control circuit.
- Established a local Wi-Fi network using Wi-Fi modules for wireless operation.
- Mapped sensor output to servo angles using forward kinematics to remotely control the robot arm and deliver first aid.

Project archive: [B.Eng. Project](#)

Additional Independent Projects

- Marker transformation function for motion-capture systems
- Path-following, forward-kinematics robot control
- 2D computer game
- K-means classification
- A* and Dijkstra pathfinding algorithms
- Reusable code library and practical learning files on HTML fundamentals
- Real-time PPG (photoplethysmography) signal display
- Gaze tracker
- Prosthetic leg design
- 2D lower-limb exoskeleton

GitHub: [elan0705](#) / Project archive: [eye_tracker, game & small projects](#)

PUBLICATIONS

Big Data and Medical Research 2020

Kannabiran, E., Prakash, T., & N., Deepa. Presented at the International Virtual Conference on Recent Trends in Electrical, Electronics, Telecommunications, Instrumentation Engineering, Medical Electronics Engineering & Physics, Session 6, 9 December 2020. Organized by professional society chapters of IEEE, IETE, ISOI, IQAC, and MHRD's IIC. A review of existing applications of big data and data-handling techniques in medical research, identifying new areas of application.

Telesurgical Bionic Arm 2019

Kannabiran, E., Banerjee, A., B.U., Goutham, & Kulkarni, S.M. International Journal of Engineering and Science Invention (IJESI), Vol. 8, Issue 11, Version 2, November 2019. e-ISSN: 2319-6734, p-ISSN: 2319-6726. A master-slave robotic arm closely matching human-arm degrees of freedom, developed for telesurgical and first-aid procedures in remote areas.

Adjustable Prosthetic Leg *In preparation*

Kannabiran, E. Manuscript in preparation; targeting submission to Frontiers in Robotics and AI or an IEEE venue. Design of an adjustable prosthetic leg that accommodates a user's physical changes over time.

CERTIFICATIONS

- Machine Learning with Python — IBM
- Data Analysis with Python — IBM
- AI Ethics — IBM
- Finance for Managers — IESE Business School